



TECHNOLOGY
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part of **HID**

TSL[®] ASCII 2 SDK – FOR ANDROID[™]

SUPPORTING DEVICES RUNNING
ANDROID 12 OR GREATER

Starting with Android 12 (API31), there is a requirement for many App permissions to be obtained from the user at run-time.

You will need to be using the [TSL® ASCII 2 Software Development Kit – for Android™ v2.5.0 or greater](#) to add support for Android 12 (or greater) devices.

To support the (API31+) requirement of *User permission checks*, **your app** will need to issue a request for *Bluetooth®* permissions where appropriate.

If this is not done, then *Bluetooth®* readers will NOT BE AVAILABLE - e.g. calls to `ReaderManager.sharedInstance().updateList()` will return an empty list.

The Android developer guide to requesting permissions is available here:

<https://developer.android.com/guide/topics/permissions/overview>

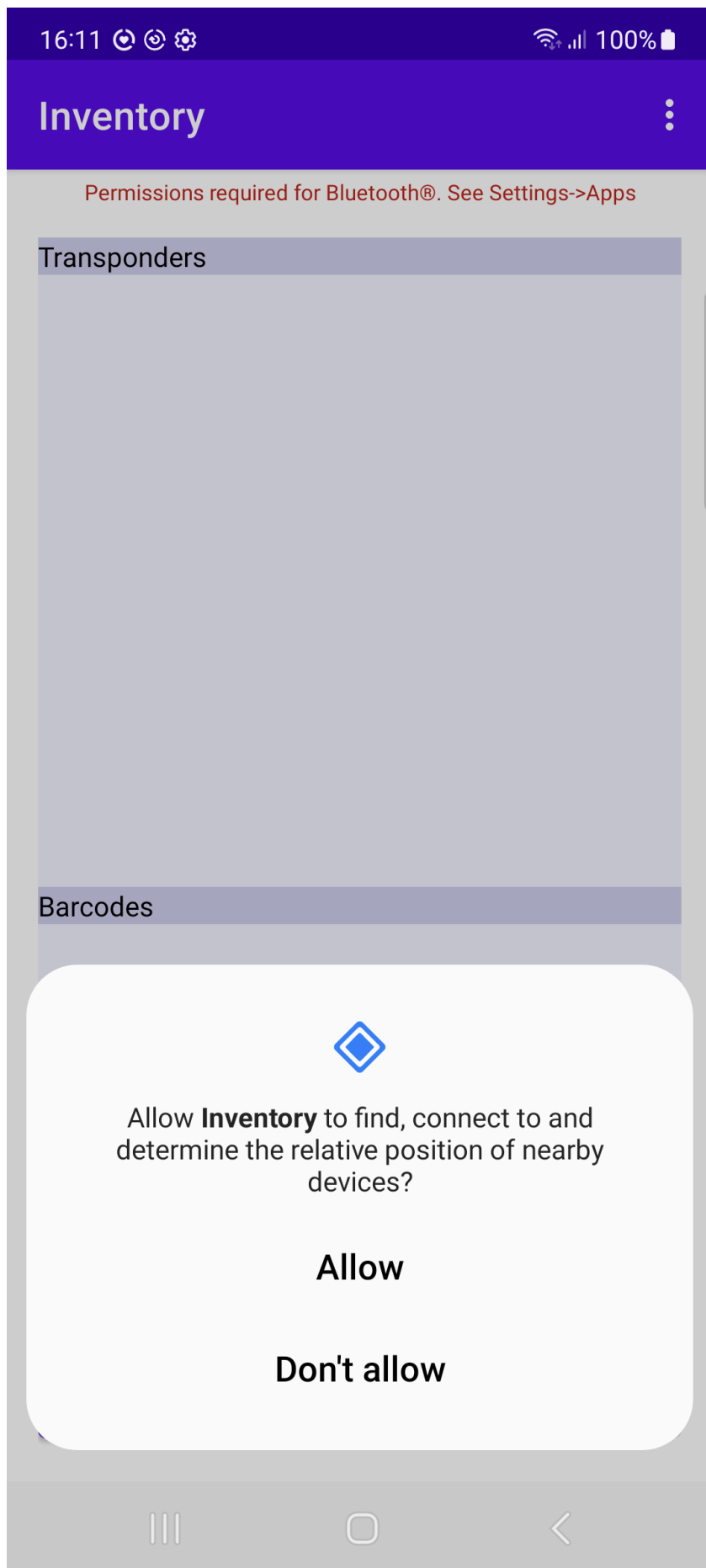
Run-Time Permissions Required

The new permissions to be requested are:

```
Manifest.permission.BLUETOOTH_CONNECT  
Manifest.permission.BLUETOOTH_SCAN
```

Note: Manifest declarations are still also required. The `Rfid.AsciiProtocol-Library` contains its own `android_manifest.xml` that will be merged with your project's `android_manifest.xml` when your project is built. The library provides both the original *Bluetooth®* permissions for older devices along with the newer permissions shown above.

At an appropriate point in your application, ask the user for the above permissions. The permissions dialog will be provided by the OS and often refers to "nearby devices" rather than mentioning *Bluetooth®* specifically. You may want to warn your users that allowing this permission is required to connect to TSL® Readers via *Bluetooth®*.



When permissions have been granted, call `ReaderManager.sharedInstance().updateList()` to ensure the reader list is up to date.

The Android SDK's sample projects have all been updated with examples of how this can be done - there are slight differences in implementation details due to the differing UIs, but the code is mostly identical.

The Sample code implementation asks for permission in the `onStart()` lifecycle method of the apps Activity, but you may find a better place in your own apps if use of the TSL® Reader is not .

```
@Override
protected void onStart()
{
    super.onStart();
    checkForBluetoothPermission();
}
```

The implementation of the `checkForBluetoothPermission()` method can be found at the end of the sample project's Activity source file.

By following the above advice, owners of the latest Android devices can continue to enjoy using your RFID App with their TSL® RFID Reader.

About TSL®



Technology Solutions UK Ltd (TSL®), part of HID Global, is a leading manufacturer of high performance mobile RFID readers used to identify and track products, assets, data or personnel.

For over two decades, TSL® has delivered innovative data capture solutions to Fortune 500 companies around the world using a global network of distributors and system integrators. Specialist in-house teams design all aspects of the finished products and software ecosystems, including electronics, firmware, application development tools, RF design and injection mould tooling.

TSL® is an ISO 9001:2015 certified company.



ISO 9001: 2015

Contact

Address:	Technology Solutions (UK) Limited, Suite A, Loughborough Technology Centre, Epinal Way, Loughborough, Leicestershire, LE11 3GE, United Kingdom.
Telephone:	+44 1509 238248
Fax:	+44 1509 214144
Email:	enquiries@tsl.com
Website:	www.tsl.com

About HID Global



HID Global powers the trusted identities of the world's people, places and things. We make it possible for people to transact safely, work productively and travel freely. Our trusted identity solutions give **people** convenient access to physical and digital **places** and connect **things** that can be identified, verified and tracked digitally. Millions of people around the world use HID products and services to navigate their everyday lives, and billions of things are connected through HID technology. We work with governments, educational institutions, hospitals, financial institutions, industrial businesses and some of the most innovative companies on the planet. Headquartered in Austin, Texas, HID Global has over 4,000 employees worldwide and operates international offices that support more than 100 countries. HID Global is an ASSA ABLOY Group brand.

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